Large-scale Bayesian inversion with complex models demands structure-exploiting high-rank Hessian approximations

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Context: Bayesian inversion provides a coherent mathematical framework for integrating data and prior information with complex models to estimate model parameters and quantify uncertainties. Efficient and robust large-scale Bayesian inversion methods open the door to tackle outer loop problems such as prediction [6], optimal experimental design under uncertainty [1], optimal control [2], and decision making under uncertainty. However, characterizing the posterior distributions for Bayesian inverse problems with high-dimensional parameters or predictions requires repeated expensive evaluations of the model and its derivatives with respect to parameters. To make solution of large-scale Bayesian inverse problems computationally tractable, it is imperative to reduce the number of such model evaluations by exploiting problem structure.

Here, we advocate exploiting problem structure to build approximations for the second derivative (Hessian) of the negative log posterior. Hessian approximations can be used, for example, to accelerate Newton-type methods for estimating the maximum a-posteriori probability (MAP) point, to build the Laplace approximation of the posterior (a local Gaussian approximation with covariance given by the inverse of the Hessian at the MAP), and Hessian-informed Markov chain Monte Carlo methods for sampling from the posterior [5]. Numerical methods that do not make use of the Hessian typically perform poorly because they have difficulty accounting for the widely varying directional scalings (ill-conditioning) of the posterior which are common in large-scale Bayesian inverse problems. The Hessian takes on even greater importance when the Bayesian inverse problem is nested within an outer loop problem. In some outer loop problem formulations the Hessian appears explicitly (e.g., prediction, optimal experimental design, optimal control), while in other cases the Hessian is central to efficient numerical methods.

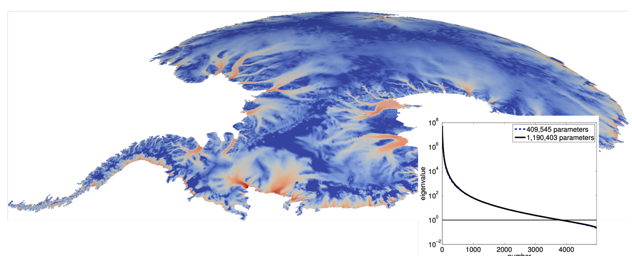

Figure 1: Surface InSAR ice flow velocity used as measurements and the spectrum of the Hessian for the ice sheet Bayesian inverse problem (inferring the basal friction coefficient). This is an example of an inverse problem with large information dimension and rich problem structure [6].

Challenges: The spectral structure of the Hessian characterizes the degree to which the data are informative about the unknown parameter: eigenvector components of the parameter corresponding to large Hessian eigenvalues are well-informed by the data, while eigenvector components corresponding to small eigenvalues are poorly informed by the data. The numerical rank of the Hessian, sometimes termed the *information dimension*, is roughly the number of components of the parameter that are informed by the data.

Low-rank Hessian approximation methods have been successfully employed for the computationally efficient solution of a wide variety of large-scale deterministic and statistical inverse problems with poorly informative data (small information dimension) by exploiting the underlying ill-posedness of the inverse problem. However, the cost to construct the low-rank approximation scales with the rank of the approximation, so the more informative the data are about the unknown parameters, the larger the computational cost [3]. In the opposite regime, stochastic data driven machine learning approaches can efficiently solve problems with extreme amounts of data. Many challenging problems of practical importance to the DOE reside in between these extremes: problems for which there is enough data that the posterior is not a low-rank perturbation from the prior, yet not enough data so that the empirical distribution can drive the numerical methods. An illustrative example is the Antarctic ice sheet inverse problem, shown in Figure 1. For these problems, new efficient high-rank Hessian approximation methods are needed.

Opportunity: At present, there is a valuable opportunity to develop efficient general purpose approximation methods for high-rank Hessians [4]. Such methods would exploit additional problem structure that applies to many Hessians of practical interest, such as locality, hierarchical low-rank structure, translation invariance, or pseudo-differential operator structure. Such methods would enable more advanced capabilities for outer loop problems with complex models as discussed above. Furthermore, in many inverse problems, one has model uncertainties/errors (due to using for instance, surrogates, or fixing parameters to nominal values) in addition to uncertainties in the inversion parameters. Addressing the challenges outlined above will enable robust parameter inversion (or solution of the outer loop problems) that accounts for such additional model uncertainties [1]. Finally, such methods would enable the integration of more complex models in real-time decision making workflows that are pertinent to application areas of interest to the DOE.

Innovation: On the methods side, developing general purpose matrix-free methods for approximating high-rank Hessians efficiently would be a breakthrough. This, in turn, would enable tackling applications that are currently not tractable. These application areas include uncertainty quantification, prediction and real time control (where applicable) for continental ice sheet dynamics, high frequency wave propagation, seismic inversion, X-ray tomography, medical imaging, estimation for electric power grids, tsunami waves, and high Reynolds number flows, among others.

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